

Step 1-2: Raw Data

dog (2s), cat (4s), lion (12s),
tiger (12.5s), mouse (20s), rat (30s)

Steps 1-5: Data Collection, Preprocessing & Embedding
Step 3: Preprocessing

Text normalization
Lowercase conversion
Punctuation removal
Stop word filtering

Step 4: Word Embeddings (spaCy)

dog: [0.12, -0.45, 0.78, ..., 0.23]
cat: [0.15, -0.42, 0.81, ..., 0.25]
300-dimensional vectors

Step 5: Vector Validation

Filter valid embeddings
Check vector norms
Remove OOV words

Step 6: Consecutive Similarities

Formula: $\cos(\theta) = \frac{\mathbf{a} \cdot \mathbf{b}}{||\mathbf{a}|| \cdot ||\mathbf{b}||}$

Steps 6-8: Similarity Computation & Phase Detection
Step 7: Phase Detection

Threshold: $\tau=0.6$ \n\nif similarity > τ :\n → Exploitation phase\nelse:\n → Exploration phase

Step 8: Detected Phases

Exploitation phase [cat]
Explore: [cat-lion]
Exploit: [lion, tiger]
Explore: [tiger-mouse]
Exploit: [mouse, rat]

Step 9: Exploitation Phase [dog, cat]

$\mu_{\text{exploit}}^{\text{intra}} = 0.82$

Steps 9-10: Intra-Phase Similarity Computation
Step 9: Exploration Phase [cat→lion]

$\mu_{\text{explore}}^{\text{intra}} = 0.38$

Step 10: Phase Centroid Calculation

Similarity Matrix:
dog cat
dog 1.00 0.82
cat 0.82 1.00

Upper triangle: 0.82

Transition similarity:
cat-lion = 0.38

For single transitions:
intra = transition similarity

For each phase:
Centroid = mean(all word vectors)

Phase 1 (dog, cat):
 $c_1 = \text{mean}([\text{dog}, \text{cat}])$
= [0.135, -0.435, 0.795, ...]

Phase 2 (lion, tiger):
 $c_2 = \text{mean}([\text{lion}, \text{tiger}])$
= [0.180, -0.380, 0.820, ...]

Step 11: Centroid Normalization

Steps 11-12: Inter-Phase Similarity Computation
Step 12: Inter-Phase Similarity

Aggregation:
 $\mu_{\text{exploit}}^{\text{intra}} = \text{mean}$
(all pairwise similarities
within exploit phases)

Inter-Phase Matrix

	P1	P2	P3
P1	1.00	.65	.58
P2	.65	1.00	.72
P3	.58	.72	1.00

Upper triangle mean: 0.65

Step 13: Coherence Ratios

Exploitation Coherence:

$= \frac{\mu_{\text{exploit}}^{\text{intra}}}{\mu_{\text{inter}}} = \frac{0.82}{0.65} = 1.26$

Exploration Coherence:

$= \frac{\mu_{\text{explore}}^{\text{intra}}}{\mu_{\text{inter}}} = \frac{0.38}{0.65} = 0.58$

Steps 13-18: Coherence Ratios & Final Metrics

Step 14: Phase Separation Index (PSI)

$\text{PSI} = \frac{\mu_{\text{exploit}}^{\text{intra}} + \mu_{\text{explore}}^{\text{intra}}}{2} - \mu_{\text{inter}}$

Step 18: Final Results

$= \frac{0.82 + 0.38}{2} - 0.65 = 0.60 - 0.65 = -0.05$

(Negative = phases less distinct)

Comprehensive metrics exported for statistical analysis

Steps 15-17: Additional Metrics

Step 15: Basic Phase Metrics

- Exploitation %: 60%
- Exploration %: 40%
- Number of switches: 4

Step 16: Clustering

- Coefficient: 0.75
- Num clusters: 3

Step 17: Word Frequency

- Mean rank: 1250
- Median rank: 980